



**COST OPTIMIZATION BASED
VEHICLE DISTRIBUTION MODEL
FOR DEALERSHIP NETWORKS**

by

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Abstract

This project focuses on optimizing Tofaş's vehicle distribution network for Fiat dealerships in Türkiye. The primary goal is to efficiently distribute various vehicle types under dynamic demand conditions while minimizing overall operating costs. The system additionally allows transfers between dealers. We created a mathematical model that solved the issue by figuring out the best routing, transfer, and inventory choices given operating constraints. The project is expected to lower costs and improve efficiency in automotive logistics systems by offering a systematic and data driven solution for distribution planning.

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Chapter 1 - Introduction

This study presents an analysis and optimization of the vehicle distribution system of Tofaş for Fiat dealerships across Türkiye. The study focuses on minimizing total operational costs in a distribution network where multiple vehicle types are delivered under dynamic demand conditions. The objective is to develop a mathematical model that shows the distribution planning process while incorporating constraints including truck capacities, dealer storage limitations, and dealer-to-dealer vehicle transfers.

The scope of this project includes identifying the parameters and decision factors required to build a mathematical model that considers operational components including dealer inventory levels, transportation decisions, and demand fulfilment processes. The developed model provides a basis for applying optimization methods.

The primary objective of the project is to achieve a balanced and efficient vehicle distribution among dealerships while minimizing total operational costs. The project is motivated by how distribution planning affects service quality and operational effectiveness, aiming to make the system more organized and economical.

To achieve these objectives, we analyzed company data, examined relevant literature, and developed solution approaches. The model determines routing, transfer, and inventory decisions under constraints. Gurobi Optimizer is used to solve the model, and heuristic techniques are incorporated to ensure efficiency in large-scale scenarios.

The mathematical model serves as a first step for solving the vehicle distribution problem and provides a foundation for future studies with more advanced optimization techniques. As a result, this project aims to improve distribution performance and support more effective decision making.

This report is divided into seven chapters. Chapter 1 introduces the project. Chapter 2 presents general information about Tofaş, Chapter 3 examines the current system and its main issues. Chapter 4 defines the problem. Chapter 5 explains the solution approach. Chapter 6 presents the results. Chapter 7 concludes the study.

Chapter 2 - Company or Public/Nonprofit Organization

TOFAŞ (Turkish Automobile Factory Inc.) is a major industrial enterprise that was founded in Turkey in 1968 by Vehbi Koç. TOFAŞ has over the years emerged as one of the major players in the Turkish automotive market with good production capacity and extensive distribution network.

The company is the representative of multiple international automotive brands such as Fiat, Alfa Romeo, Jeep, Citroen, DS Automobiles, Opel, and Peugeot, and the company produces passenger vehicles and light commercial vehicles.

TOFAŞ has its primary production plant in Bursa where manufacturing, assembly, logistics and quality control processes are combined. The Istanbul headquarters helps in coordination of production and distribution planning.

It has a workforce of approximately 7,500 employees, and its yearly production is about 450,000 vehicles. With such a scale, it is necessary to conduct efficient distribution planning that would lead to timely delivery and cost reduction. TOFAŞ is structured into main functional units, which include the production, logistics, supply chain management, quality control, human resource, finance, and IT. The specific aspects of this project are especially concerned with logistics and supply chain management that is in charge of distributing vehicles.

TOFAŞ operates in local and international markets, exporting much of its production and having a strong domestic presence with an extensive network of dealers. In this project, the interest is on Fiat brand which has 71 dealerships in Turkey and the vehicles are distributed to the dealerships out of the Bursa plant.

The distribution system is a complex decision making issue due to the availability of different models of vehicles, the requirement of different dealerships, and alternative transportation having low capacities. Therefore, the first objective of the project is to develop an efficient distribution plan that establishes how vehicles will be distributed and shipped within a day long period and at the same time reduce the total cost of operation and transportation and good utilization of resources.

Chapter 3 - Current Situation and the Problem

The current system involves distributing vehicles produced at the Bursa factory to 71 dealerships across Türkiye, divided into 7 designated regions within the company. The TOFAŞ Bursa factory produces four different vehicle types belonging to the Fiat group (Egea Sedan, Egea Cross, Scudo, and Ulysse). During distribution, a truck can only carry one vehicle type, but it is not required to carry the same type throughout the year. Once the route is completed, the truck can be adjusted to carry a different type of vehicle. Because of this, the distribution process is more complex compared to classic vehicle routing problems.

Over time, the system has become more complex, mainly due to the pandemic and its aftermath, where demand consistently exceeded supply. Dealerships always sold the vehicles allocated to them, often with even more demand. However, production is operating at full capacity. Therefore, dealer targets and sales pressure make distribution decisions more critical. Consequently, the current system is not just a route planning issue, but a multi-dimensional decision problem that requires considering capacity, time, and product variety together.

A significant part of the problem in the system is that each component directly or indirectly affects the others. The projected number of vehicle sales and shipments affects dealer inventory levels, which in turn affects customer demand. Insufficient or unbalanced distribution can create shortages in some dealers and surpluses in others. It also directly affects route length and delivery times. Unnecessary route length increases costs, while delivery delays reduce customer satisfaction. In addition, since trucks are leased annually, determining the correct number is critical. Using too many trucks increases costs, while too few leads to delays. Dealer to dealer transportation within the same region can reduce cost due to short distances, but creates additional operational complexity if the right vehicle type is not available.

Production capacity is operating at full capacity, so a more optimal distribution plan should not be seen as simply increasing sales. Since demand is consistently higher than supply, demand forecasting becomes the critical decision. Headquarters sets targets based on historical data and seasonality, but these are only forecasts and do not reflect actual demand. Therefore, the problem shifts from a “meet demand” approach to making the best use of limited resources and minimizing cost.

The system contains a number of conflicting goals. There must be a balance between service level and cost minimization. Reduced routes and optimization minimize costs, but may cause delays for some dealers. Another trade-off is between delivery speed and planning efficiency. Increasing shipment frequency improves dealer satisfaction but raises operational costs. In addition, dealers can sell cars that have not yet arrived. This can boost short-term sales performance but may lead to customer dissatisfaction due to delays. Therefore, the system should create a balance between operational efficiency and customer experience.

The most notable expectation of dealers is faster and more predictable delivery. Constant delays complicate sales planning and negatively affect customer satisfaction. They also expect a more balanced and transparent allocation of vehicles. Since supply is lower than demand, this increases competition and operational pressure among dealers. Along with service quality, cost efficiency is also a key expectation. The system should reduce logistics costs while maintaining a fair and reliable distribution structure.

The underlying causes of the issue can be summarized into a few categories. First, demand is consistently higher than supply. Second, there is limited flexibility due to fixed production capacity. Third, operational constraints, such as loading trucks with a single vehicle type, increase planning complexity. In addition, uncertainty in deliveries and the need to reduce costs make the problem more critical. Environmental impacts, such as carbon emissions, also increase the importance of cost optimization.

The existing system uses data such as demand forecasts, distances, transportation costs, truck capacity, and dealer inventory levels. However, a holistic optimization approach is not effectively utilized. Planning is mostly based on predetermined targets rather than fully optimized operational decisions. This leads to unnecessarily high logistics costs and inconsistent delivery performance. As a result, the current system is insufficient in terms of both customer satisfaction and cost effectiveness, and a more balanced, predictive, and optimized distribution system is needed.

Chapter 4 - Problem Description

4.1) Problem Definition

This project focuses on the design of a complex vehicle distribution system that optimizes the allocation of final vehicles from the manufacturing facility to authorized dealers while considering certain limitations and observable problems. This is a physical and decision-making system. It includes Automotive supply chain, vehicle distribution network, demand forecasting and decision support system.

We developed a system that supports Tofas which improves the existing vehicle distribution plan by integrating service level constraints, demand uncertainty and CO₂ emission effect. Also we did not only build a model, but also analyzed a real industrial system and developed an engineering solution. Automotive manufacturers, under time and cost constraints, are forced to distribute large quantities of finished vehicles to geographically dispersed dealerships. In the case of Tofas, the vehicles produced at the manufacturing plant in Bursa need to be delivered to dealerships across Türkiye; this leads to a complex nationwide distribution problem.

The model requires several key inputs. On the demand side, we give demand inputs considering the forecasted numbers of demands. On the supply side, we have limited production capacity and a certain number of trucks annually, so these data are given as inputs. In terms of transportation, we set distances between dealerships (nodes) as we collected data from the company, and it also decided a cost per km and the emission factors in terms of trucks capabilities. For inventory, current stock levels of dealerships and capacity limit of factory and trucks are given beforehand.

The outputs of the proposed system include the allocation of vehicles to specific dealers, followed routes with planning decisions by each truck and an overall distribution plan. The model also provides key performance indicators such as total operating cost and service level, which are used to evaluate the effectiveness of the solution.

The system consists of the dealers as demand nodes, factory as production and inventory, trucks as transportation resources which are supported by the Transportation Network (roads). Additionally, the information system (data flow) manages data including demand, inventory levels, and transportation parameters to support decision-making.

4.2) Problem formulation

Several assumptions are conducted throughout our model. Each truck is assigned to a specific car type during its route. Also they have a fixed capacity regarding the assigned car type. Travel times between nodes are deterministic and known in advance. Moreover, the demands are represented using forecasting values which we will be estimating. Inventory levels are tracked daily of which data collected from the company. In addition, we excluded the factors that are difficult to observe and predict such as traffic, breakdown...

The aim of the model is to minimize the total operational cost throughout the planning and the transportation period. The total cost includes transportation costs, inventory holding costs, daily order delay penalties, and penalties applied to unsold vehicles by the end of the planning period. Additionally, a small penalty is applied for tour time to encourage more efficient route planning decisions. This multi-component objective function supports not only operational efficiency but also service performance.

The model includes several constraints. In terms of capacity and loading, even though each truck has a fixed capacity, it can vary its capacity according to the assigned car type and these constraints ensure that the number of loaded vehicles on each truck does not exceed its capacity. For routing, these constraints ensure that each truck starts and ends its tour at the factory, while guaranteeing the feasibility of routes, maintaining flow conservation, and avoiding subtours. Regarding scheduling constraints, routes are prevented from overlapping through a locking mechanism that keeps trucks unavailable during ongoing tours.

The performance of the system is evaluated using key indicators such as total cost, service level, backorder levels, inventory utilization, and truck utilization. While the model captures the fundamental dynamics of the distribution system, some real-world uncertainties, such as unexpected delays, traffic congestion and operational disruptions are not explicitly included. Moreover, demand is based on forecasts and may differ from actual values. Due to the complexity of the integrated model, solving large-scale samples using exact optimization methods becomes computationally costly. Therefore, heuristic approaches based on demands and routing are also considered to obtain high-quality solutions within reasonable computation times.

4.3) Physical, Technical, Social, Environmental and Economic Constraints

This issue is fraught with a number of limitations. Operation is limited by physical factors of truck capacity and dealer storage. Mathematically, the complexity of models and the constraint of data influence the quality of solutions. In social terms, it is vital to have an equitable distribution among dealers. Transportation affects the environment negatively by creating carbon emission. Transport, inventory, and fleet size are some of the costs that need to be kept down economically without reducing the level of services.

4.4) Related Industrial Engineering Courses

Industrial Engineering Courses	Where is it used?
IE 201 Operations Research/ IE 301 Operations Research II / IE 342 Mathematical Modeling and Exact Methods	Formulation/ Design of Mathematical Model
IE 246 Programming for Operations Research	Using Gurobi for solving the mathematical model
IE 343 Mathematical Modeling and Heuristic Methods	Subtour Elimination
IE 366 Heuristic Methods for Optimization	Identifying the VRP Model
IE 383 Decision Support Systems	Applying Excel Pivot Tables and Statistical Analysis Tools

This table illustrates the connection between theoretical knowledge from Industrial Engineering courses and its practical implementation in the project, highlighting how different methods and tools from lectures contribute to solving real world problems.

4.5) Related United Nations SDG's

Goal ID	Goal	Tick if related to your project	Explanation
12	Responsible Consumption and Production	x	Efficient resource use
13	Climate Action	x	Emission reduction

SDG 13, Climate Action: While our main goal started out as figuring out how to cut transportation costs, we realized that the best way to do that is through smarter routing. By optimizing how vehicles move, we are actively keeping them on the road for less time and burning less fuel. This directly translates to a meaningful drop in carbon emissions.

Chapter 5 - Design and Solution Approach

5.1) Literature Review

i) Inventory Routing Problem

Inventory routing problem is one of those optimization issues receiving extensive research. The problem involves integration of two logistics processes into one inventory management and routing. The difference between classical logistic problems and IRP is that the latter includes delivery timing, shipping volumes and route optimization to decrease logistics costs altogether [1]. Classical IRP models have been created based on unrealistic assumptions like deterministic demand, fixed transport costs and stable network configuration. However in reality logistic operations occur within more complex environments. Thus modern researchers pay particular attention to making IRP more realistic through introduction of such factors as cost dependency on time, stochastic demand, multi-period plans, and multi-product logistics [2]. These extensions provide more opportunities for application of IRP in problems in the real world. The primary advantage of IRP is reduction of supply chain logistics cost by integration of decisions regarding inventory management and transportation processes. As opposed to optimization of every single aspect, integration helps to achieve higher levels of efficiency and better usage of resources [3]. In addition some researchers take into account the uncertainty of supply and demand, which is particularly important for highly dynamic industrial logistics [4]. In this case, considering the distribution problem in the automotive sector, IRP acquires additional significance because of the changing demand conditions and the complicated nature of the operations. In particular, the automotive industry comprises different types of vehicles, widely distributed dealerships, and tight schedules, making the use of an integrative approach important for addressing the issues arising [5]. Additionally according to contemporary research on the topic IRP can incorporate not only production but procurement processes as well. This means that the whole system should be considered to obtain the best results[6]. However despite its benefits IRP is a computationally difficult task. In fact, exact approaches to finding an optimum such as MILP, can be applied to problems on a small scale, while heuristic and meta-heuristic techniques are better suited for problems that have a larger scale [2].

ii) Demand Forecasting

Demand forecasting is an integral aspect of automotive logistics that is vital in determining production planning, inventory and distribution. Precise forecasting can help businesses balance demand with supply to prevent problems like shortages or excess inventory. From available literature, forecasting can be defined as the act of making predictions about the future state of demand by using historical records and past trends [7]. Demand within automotive logistics is quite volatile due to the presence of numerous external factors such as the economy and seasonality [8]. Consequently forecasting becomes crucial within automotive logistics. Some of the common methods of forecasting in practice include Least Squares, Single Exponential Smoothing and Double Exponential Smoothing [7]. These traditional forecasting models are simple, reliable and easy to understand. However, among these three models, Double Exponential Smoothing is effective since it provides accurate forecasts in scenarios where there are trends in the data, with minimal errors like Mean Absolute Percentage Error of 3% [7]. Nevertheless, modern research indicates that classical models of statistical analysis may not be sufficient in many situations. ARIMA models, machine learning models and other sophisticated methods of prediction are being applied more frequently. They allow incorporating external factors and analyzing nonlinear correlations between different demand factors [8]. The literature indicates that the choice of forecasting models depends not only on the selection of the most suitable technique for prediction but also on the analysis of the demand pattern. The automobile demand may be characterized by the presence of trends, seasonality and random fluctuations, necessitating the choice of suitable models according to the features of the analyzed dataset [7].

iii) CO₂ Emissions and Sustainability in Logistics

Sustainability has become a key concern in logistics due to increasing environmental awareness. Transportation is responsible for a significant share of global CO₂ emissions, particularly in road transport [11]. Recent studies emphasize integrating environmental factors directly into optimization models rather than considering them separately [10]. Emission aware routing models account for factors such as vehicle type, speed and traffic conditions, which significantly affect emissions [9]. Additionally, integrated supply chain models now aim to minimize both costs and CO₂ emissions simultaneously [10].

5.2) Modeling Approach and Justification

In this research, a MILP formulation will be employed as the model to solve the given problem. The major advantage of employing MILP in solving this problem arises from the minimization of the overall system cost (transportation cost, inventory cost, backlogging cost and penalty cost). This is due to the multi-period nature, multi-product structure and combination of inventory and routing decisions subject to capacity constraints involved in the problem. Besides, MILP allows an exact mathematical modeling of the system since the decision variables, such as routing decisions, inventory level decisions and shipment decisions together with constraints of the problem can be clearly expressed. In other words, this makes it possible to have a structured representation of the given system. The model which is formulated in this project is referred to as the IRP with the consideration of multiple periods and multiple products. The model incorporates various types of decision making factors, which include routing decisions, inventory decisions and loading decisions as well as capacity and flow constraints. Time-dependent demands are considered in place of constant demands. Finally, total system cost is optimized using the formulated model. MILP is one of the most common optimization techniques applied to solve the problem of IRP and gives the most accurate results even for small and medium sized instances. At the same time, MILP suffers from some limitations. One of the main drawbacks of using MILP is that it requires a considerable amount of computing resources, because the number of variables and constraints grows proportionally to the growth of the instance. Another drawback of MILP is that the cost functions and other input values are considered to be deterministic and constant within the solution procedure. Thus, MILP may not be an efficient technique for solving large and dynamic problems. In particular, the problem of IRP may need to be solved in real time or with the use of more complex data, then MILP will not be the optimal choice of the optimization method. Instead it would be more effective to apply heuristics or some other optimization techniques.

5.3) Representation of the Models

i) Mathematical Model

1. Sets (Indices)

- C : Set of car models, indexed by c .
- K : Set of trucks in the fleet, indexed by k .
- N : Set of all nodes, indexed by i, j (node 0 = Factory).
- T : Set of time periods (days), indexed by t .

2. Parameters

- A_{ij} : Travel time from node i to node j .
- B_c : Backorder penalty cost per unit of car type c per day.
- F_{ict} : Forecasted demand for car type c at node $i \in N \setminus \{0\}$ on day t .
- G_c : Penalty cost per unsold car of type c at the end of the month.
- H_c : Holding cost per unit of car type c per day.
- M : Sufficiently large positive number, Big M.
- P_t : Binary reset switch; 1 if day t is the first day of a new month, 0 otherwise.
- Q_c : Maximum truck capacity when loaded with car type c .
- R_{ij} : Transportation cost to drive from node i to node j .
- V_i : Maximum vehicle storage capacity at node $i \in N \setminus \{0\}$.

3. Decision Variables

Routing & Tour Time:

- $X_{kijt} \in \{0, 1\}$: 1 if truck k travels from i to j on a tour dispatched on day t .
- $Y_{kt} \in \{0, 1\}$: 1 if truck k starts a tour on day t .
- $W_{kt} \in \mathbb{Z}_{\geq 0}$: The time the truck k 's specific tour takes.
- $L_{kt} \geq 0$: The number of days truck k is currently locked/unavailable on day t .
- $Z_{kct} \in \{0, 1\}$: 1 if truck k is assigned to carry car type c on the tour starting day t .

Load & Operations:

- $J_{kijt} \geq 0$: Physical number of cars on truck k while driving from i to j on tour t .
- $D_{kict} \geq 0$: Cars dropped off at node i by truck k on tour t .
- $U_{kict} \geq 0$: Cars picked up at node i by truck k on tour t .

Inventory:

- $I_{ict} \geq 0$: Daily inventory level of car type c at node i on day t .
- $E_{ict} \geq 0$: Daily backorder level (unmet demand) for car type c at node i on day t .
- $S_{ict} \geq 0$: Unsold cars at node i for car type c on day t .

4. Objective Function

Minimize total costs. A very small penalty (ϵ) is applied to W_{kt} to ensure the solver selects the smallest integer tour time.

$$\begin{aligned}
\min \quad & \underbrace{\sum_{k \in K} \sum_{i \in N} \sum_{j \in N} \sum_{t \in T} R_{ij} X_{kijt}}_{\text{Transportation Cost}} + \underbrace{\sum_{i \in N \setminus \{0\}} \sum_{c \in C} \sum_{t \in T} H_c I_{ict}}_{\text{Holding Cost}} + \underbrace{\sum_{i \in N \setminus \{0\}} \sum_{c \in C} \sum_{t \in T} B_c E_{ict}}_{\text{Daily Backorder Cost}} \\
& + \underbrace{\sum_{i \in N \setminus \{0\}} \sum_{c \in C} \sum_{t \in T} G_c S_{ict}}_{\text{Unsold Car Cost}} + \underbrace{\epsilon \sum_{k \in K} \sum_{t \in T} W_{kt}}_{\text{Tour Time Penalty}}
\end{aligned} \tag{1}$$

5. Constraints

A. Routing Flow

Let u_{kit} be an auxiliary continuous variable representing the sequence position of node i in truck k 's tour on day t .

$$\sum_{j \in N} X_{k0jt} = Y_{kt} \quad \forall k \in K, t \in T \tag{2}$$

$$\sum_{i \in N} X_{ki0t} = Y_{kt} \quad \forall k \in K, t \in T \tag{3}$$

$$\sum_{j \in N} X_{kjit} = \sum_{j \in N} X_{kijt} \quad \forall k \in K, i \in N, t \in T \tag{4}$$

$$\sum_{c \in C} (D_{kict} + U_{kict}) \leq M \sum_{j \in N} X_{kjit} \quad \forall k \in K, i, j \in N, t \in T \tag{5}$$

$$u_{kit} - u_{kjt} + |N| \cdot X_{kijt} \leq |N| - 1 \quad \forall k \in K, t \in T, i, j \in N \setminus \{0\} : i \neq j \tag{6}$$

B. Truck Capacity and Load Conservation

$$\sum_{c \in C} Z_{kct} \leq Y_{kt} \quad \forall k \in K, t \in T \tag{7}$$

$$D_{kict} + U_{kict} \leq Q_c Z_{kct} \quad \forall k \in K, i \in N \setminus \{0\}, c \in C, t \in T \tag{8}$$

$$J_{kijt} \leq M \cdot X_{kijt} \quad \forall k \in K, i \in N, j \in N, t \in T \tag{9}$$

$$J_{kijt} \leq \sum_{c \in C} Q_c Z_{kct} \quad \forall k \in K, i \in N, j \in N, t \in T \tag{10}$$

$$\sum_{j \in N} J_{kjit} + \sum_{c \in C} U_{kict} - \sum_{c \in C} D_{kict} = \sum_{j \in N} J_{kijt} \quad \forall k \in K, i \in N \setminus \{0\}, t \in T \tag{11}$$

$$\sum_{j \in N} J_{kj0t} = 0 \quad \forall k \in K, t \in T \tag{12}$$

C. Route Time Tracking & No Overlapping Tours

$$\sum_{i \in N} \sum_{j \in N} A_{ij} X_{kijt} \leq W_{kt} \quad \forall k \in K, t \in T \quad (13)$$

$$L_{kt} \geq L_{k,(t-1)} - 1 + W_{k,(t-1)} \quad \forall k \in K, t \in T \quad (14)$$

$$L_{kt} \leq M(1 - Y_{kt}) \quad \forall k \in K, t \in T \quad (15)$$

D. Daily Dealer Inventory, Backorders, and Monthly Reset

$$I_{ict} - E_{ict} = (1 - P_t)(I_{ic,(t-1)} - E_{ic,(t-1)}) + \sum_{k \in K} D_{kict} - \sum_{k \in K} U_{kict} - F_{ict} \quad \forall i \in N \setminus \{0\}, c \in C, t \in T \quad (16)$$

$$S_{ic,(t-1)} = P_t \cdot I_{ic,(t-1)} \quad \forall i \in N \setminus \{0\}, c \in C, t \in T \quad (17)$$

$$\sum_{c \in C} I_{ict} \leq V_i \quad \forall i \in N \setminus \{0\}, t \in T \quad (18)$$

Objective Function and Constraint Explanations: (1) The objective function, which minimizes total operation costs in the total time period. (2) If a truck is dispatched on a given day, it must leave the factory exactly once. (3) If a truck is dispatched on a given day, it must return to the factory exactly once. (4) Routing flow conservation. If a truck enters a node, it must also depart from that node. (5) A truck cannot pick up or drop off cars at a node unless it actually travels to that node. (6) Subtour elimination to ensure all truck routes form valid paths connected to the factory. (7) A dispatched truck can only be assigned to carry only one specific car type for its entire tour. (8) A truck only picks up or drops off the specific car type it was assigned to carry. (9) A truck cannot carry any cars on an arc it does not traverse. (10) The number of cars on a truck cannot exceed its maximum physical capacity limit. (11) Balance equation for the number of cars on the truck as it travels between nodes. (12) All trucks need to be completely empty of inventory when returning to the factory. (13) The total travel time required for a truck's tour. (14) The locking mechanism to keep a truck unavailable for the duration of its previous tour. (15) A truck cannot dispatch on a new tour if it is still locked from a previous trip. (16) Daily inventory and backorder level at each dealer. (17) The trigger for the unsold cars at the end of the month. (18) The total inventory of all car types held at a dealer never exceeds its storage capacity.

ii) Heuristic Solution- Greedy Algorithm

The heuristic algorithm below is designed to quickly generate feasible vehicle routing schedules for large instances (like our original problem) where the exact solution is impossible to compute.

The algorithm takes the transportation network parameters (travel times, costs), truck constraints (capacities, availability), and daily forecasted dealer demands. The heuristic operates day by day. For each day, it iterates through each car type sequentially. It identifies "critical dealers" whose current inventory is insufficient to meet the day's forecasted demand.

If available trucks and critical dealers exist, a truck is dispatched. The route is constructed using a "nearest neighbor" greedy approach. The truck searches for all critical dealers sorted by travel time from its current location and moves to the closest one. It delivers the required quantity to satisfy the demand of that dealer. If a truck satisfies a dealer's demand but still retains inventory, it continues to the next nearest critical dealer. If no critical dealers remain but the truck still holds inventory, the truck delivers the remaining vehicles to the final dealer on the route to avoid returning inventory to the factory.

As output, it gives daily satisfied demand percentage including total demanded and delivered quantities, first 30 routes for verification, and end of the month report (total unmet demand, or unsold cars for each car type). For the entire pseudocode, please check "Appendix" (Figure 1.1).

5.4) Verification and validation of the Exact model

For the verification of the model, a Gurobi solver was applied to an instance of 10 nodes, 5 trucks, and 30 days. Within the 600 second computational time limit, the solver explored 7321 nodes and generated 10 feasible solutions, with a best objective value of 70612 (for details, use the GitHub link to check the entire code in the "Appendix" section). The console output confirms that the model correctly respected all bounds; for instance, unsold cars at the end of the month were minimized to 0.0 (given a large penalty for unsold cars), and truck flows were balanced.

```
Explored 7321 nodes (1805684 simplex iterations) in 600.11 seconds (477.00 work units)
Thread count was 16 (of 16 available processors)

Solution count 10: 70612.1 70639.7 70639.7 ... 70916.3

Time limit reached
Best objective 7.061211400000e+04, best bound 3.778680615526e+04, gap 46.4868%

RESULTS
Obj Value: 70612.114
Unsold Cars: 0.0
```

Figure 2.1: *Console Outputs of the Gurobi Solver*

By visualizing the daily dispatch schedules (Figures 2.1 to 2.4, representing Days 0 through 3), we can confirm the logical behavior of the model. On Day 0, Truck 1 (Skudo) dispatches from factory to Dealer 1, Truck 2 (Egea) to Dealer 3, and Truck 4 (Egea) to Dealer 4. The model accurately calculates travel times: T1 and T4 take 3 days for their determined tours. As observed in the Day 3 plot, T1 and T4 successfully return to the factory, unlock, and T1 immediately restarted routing to new dealers (Dealer 8 and Dealer 10). The exact model effectively calculates the shortest paths and completely eliminates overlapping or subtours. With the help of the plotting results below, we can prove the exact model's validity for small scale problems.

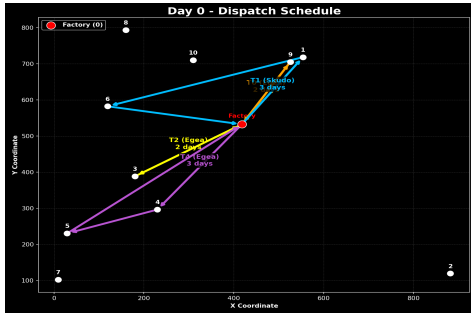


Figure 3.1: Routing Schedule on Day 0

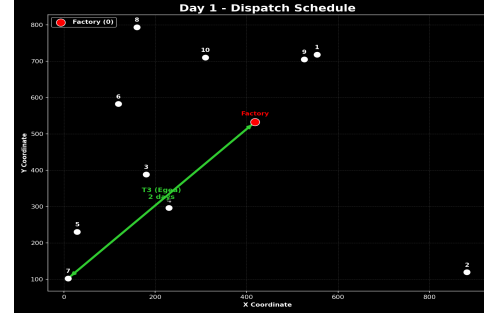


Figure 3.2: Routing Schedule on Day 1

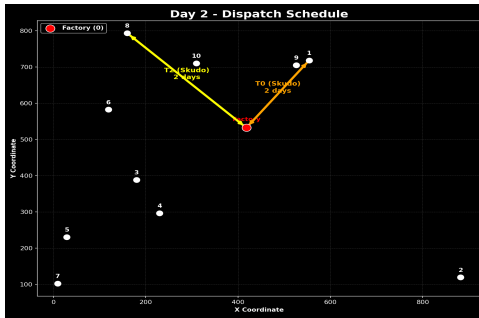


Figure 3.3: Routing Schedule on Day 2

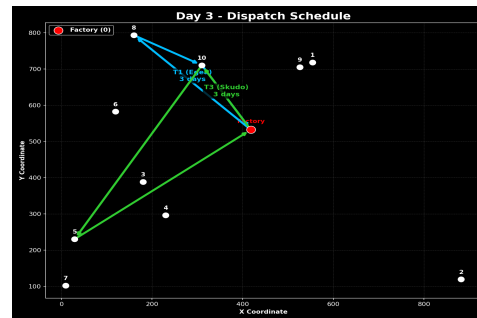


Figure 3.4: Routing Schedule on Day 3

5.5) Verification and validation of the Heuristic model

For the verification and the validation of the heuristic model, the logic is implemented to a Python script, and the algorithm was applied to the exactly same problem (10 nodes, 5 trucks, 30 days instance). Also as seen in the plots given in the “Appendix” section (from Figure 4.1 to 4.4), the algorithm tends to go to the nearest node and builds up its road according to that logic. Also, by using the GitHub link in the “Appendix”, you can check all 30 days’ plots for both exact and heuristic algorithms.

Chapter 6 - Computational Results

In this chapter, the computational experiments conducted to evaluate the performance, scalability, and parameter analysis of the models are provided. The experiments are divided into two main phases: an evaluation of the heuristic algorithm over a 30 day period, and a sensitivity analysis of the exact model to observe its behavior under varying parameters. All exact models were solved using Gurobi Optimizer in Python with a strict computational time limit of 600 seconds. The whole script is available on GitHub, and is mentioned in the “Appendix” section.

6.1) Results of computational experiments

To evaluate the mathematical and computational behavior of the proposed models, we conducted sensitivity analyses on both the exact model and the heuristic algorithm. These tests focus on how the network responds to varying parameters.

i) Exact Model Sensitivity to Backorder Penalties

First, we analyzed the exact model's response to customer service requirements by varying the backorder penalty parameter. The configuration we used had 5 dealers, 10 trucks, 7 days period, 2 car types, and the baseline penalty was scaled by factors of 0.5, 1.0, 2.0, and 5.0.

The results show a clear trade off. When the penalty is low (0.5x), the solver is willing to accept 3 backordered units to save on transportation, keeping transit costs down to 15,226.71. However, at standard (1.0x) and extreme (5.0x) penalties, the model enforces a perfect service level (0 unmet units), forcing more aggressive routing and driving transportation costs above 16,000 and 17,000, respectively (for the entire table, please check the “Appendix” section Table 1.1).

ii) Exact Model Sensitivity to Truck Numbers and Objective Value

Next, we evaluated the exact model with the same configuration against different truck quantities (4, 6, and 8 trucks). Normally the MILP model scales exponentially, but we restricted the computation limit with 600 seconds.

In the results, for the truck numbers of 4, 6, 8; the objective function value greatly decreases to 29742, 19132, and 18157 respectively (for the full table, check Table 2.1 in “Appendix”). With 4

trucks, the solver fails to satisfy all demands, resulting in a huge backorder cost of 12500. So, an increase in the number of trucks helps us achieve a better objective value.

iii) Heuristic Model Sensitivity to Truck Numbers and Objective Value

To compare with the exact model, the heuristic algorithm was tested across the same truck numbered rs. Unlike the exact model, the heuristic scales perfectly in terms of speed, solving all scenarios in under 0.001 seconds. Similar to the exact model, shifting from 4 to 6 trucks eliminates backorders. However, the heuristic's aggressive routing behavior leads to much higher unsold inventory costs as the number of trucks increases (which leads to a higher total costs, unexpectedly), since each truck drops the remaining cars to the last node of the route (see the heuristic pseudocode in “Appendix”, figure 1.1). Also, the full table is available Table 2.2 in the “Appendix” section.

6.2) Interpretation and discussion of the results, proposed solution, design alternatives

The exact algorithm is not applicable to the original problem since it is a lot more complicated than the trial run. For this reason, the only possible solution method is heuristic. The current heuristic algorithm is designed to be simple and easily applicable. As expected, simplicity comes with a cost.

As seen in the heuristic solution's outputs, it gives outputs which are far away from the optimal solution. The main reason behind this is, as mentioned at the previous section too, that in this algorithm each truck drops all the remaining cars to the last node of the route. This situation causes a lot of cars to stay unsold at the end of the month (also see the pilot run results).

Another problem is that this algorithm iterates over each car type one by one. Meaning that the algorithm may start using trucks for the next car type when it's done for the current car type. This situation causes the last car types' demands to remain unmet, unlike first car types (also see the pilot run results).

To solve those 2 main problems, we'll try different design alternatives. While having that greedy logic, the unsold car problem and imbalance between car type satisfactions can be easily overcome by different design techniques.

Further information about our next steps at this point and design alternatives are also provided at the “Conclusion” section.

6.3) Confirmation of computational results on pilot implementation

For pilot implementation, we run the heuristic algorithm, onto a randomly generated data which has as many variables as the original problem (72 dealers, 5 car types, 1000 trucks, 1 month period). The console outputs are as follows:

```

END OF MONTH INVENTORY REPORT
=====
Car Type 0: Unmet Demand = 13 | Unsold = 5624
Car Type 1: Unmet Demand = 0 | Unsold = 25895
Car Type 2: Unmet Demand = 0 | Unsold = 5403
Car Type 3: Unmet Demand = 0 | Unsold = 17348
Car Type 4: Unmet Demand = 8847 | Unsold = 1757
-----
Unmet Demand = 8860, Unsold Cars = 56027
=====

```

Figure 5.1: Console Outputs of the Heuristic Pilot Run

Day 4				Day 29			
Car Type 0:	Demanded = 438	Delivered = 397	Satisfaction = 90.6%	Car Type 0:	Demanded = 427	Delivered = 428	Satisfaction = 100.2%
Car Type 1:	Demanded = 454	Delivered = 450	Satisfaction = 99.1%	Car Type 1:	Demanded = 397	Delivered = 405	Satisfaction = 102.0%
Car Type 2:	Demanded = 430	Delivered = 1078	Satisfaction = 250.7%	Car Type 2:	Demanded = 426	Delivered = 450	Satisfaction = 105.6%
Car Type 3:	Demanded = 431	Delivered = 0	Satisfaction = 0.0%	Car Type 3:	Demanded = 382	Delivered = 352	Satisfaction = 92.1%
Car Type 4:	Demanded = 426	Delivered = 0	Satisfaction = 0.0%	Car Type 4:	Demanded = 423	Delivered = 125	Satisfaction = 29.6%

Figure 5.2: Day 4 Outputs of Pilot Implementation Figure 5.3: Day 29 Outputs of Pilot Implementation

In this pilot implementation, problems stated beforeward (unsold cars, and imbalance) are clearly visible in the outputs above. In Figure 5.1, the number of total unsold cars is overwhelming (especially having a huge penalty cost per unsold cars). The reason for having that many unsold cars can be easily seen in Figure 5.2, which states a delivered/ demanded car ratio of 250%. And in both Figure 5.2 and 5.3, as we move on to the next car type (especially for car type 3 and 4), the delivered amount drastically decreases, which again signals a need for a balance between each car type deliveries.

Chapter 7 - Implementation

Model deployment involves the regular collection and integration of dealer demand, inventory levels, and transportation data within the system. This process is expected to be integrated with the models provided.

To monitor system performance, indicators such as total cost, delivery time and vehicle utilization rates should be tracked. Model results should be compared with actual operational performance and any insufficiencies should be addressed accordingly.

The implementation also requires appropriate software infrastructure, a Gurobi license and basic personnel training to effectively use the system. While the exact model is technically powerful, solving large scale problems is not possible because of computation times. On the other hand, the heuristic algorithm gives solutions in milliseconds but this model lacks accuracy. Although there will be initial financial costs and long term savings are expected.

Despite this study including a full scale implementation, the heuristic algorithm needs critical improvements. Further development will be made in IE 402.

Chapter 8 - Conclusion

In this study, we examined the inventory distribution system for Fiat dealerships in Türkiye, based on the problems identified by Tofaş and developed a mathematical model to address this system in subsequent stages. The model considers all relevant constraints, addressing vehicle distribution, dealership inventories and transfers between dealers. Its functionality was tested using Gurobi on a small scale instance (compared to the real problem) and using a heuristic algorithm on a large instance (same settings with real data).

While the primary objective of this study is to minimize costs, contributing to environmental and social impacts is also a key priority. More efficient routing supports the reduction of carbon emissions while maintaining a lower transportation cost improves system quality.

Data privacy and fairness among dealerships are key ethical considerations. The mathematical model is designed to ensure that no dealership gains an unfair advantage and the decision making process is made only according to the cost minimization.

The current work is limited to model development and exact & heuristic solutions, both on small and large datasets. In future work, the current heuristic model will be enhanced according to the problems mentioned (unsold car costs and imbalance) by testing different design techniques (carrying limited cars for unsold cars problem, a dynamic selection between car types for imbalance problem). After doing so, meta heuristic methods will be developed to obtain more effective solutions by destroying and reconducting the gathered solutions from the base heuristic algorithm (like genetic algorithm).

References

In this part, there should be a list of references and each reference has its own number in the reference list. There may be several types of references like web pages, books, articles or conference proceedings. As mentioned in the Format of Project Section on page 3, refer to IEEE documentation at <http://www.ieee.org/documents/ieeecitationref.pdf>.

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Appendices

GitHub repository including the entire code: <https://github.com/muratorkunkatkat/IRPSenior>

(Note that in this repository, all the plotting outputs are also available.)

Heuristic Algorithm

Algorithm 1 Greedy Insertion Heuristic

```
1: for period  $t$  (Days)  $\in T$  do
2:   for car type  $c \in C$  do
3:     for dealer  $i \in N \setminus \{0\}$  do
4:       Check if dealer  $i$  is critical:  $(I_{ict} - F_{ict}) < 0$ 
5:     end for
6:     while (available trucks  $> 0$ ) and (critical dealers for  $c > 0$ ) do
7:        $current = 0$  (Factory)
8:       Pick an available truck  $k \in K$ 
9:        $available\_capacity = Q_c$  (given that car type is  $c$ )
10:       $route = []$ 
11:       $route\_time = 0$ 
12:      while (available_capacity  $> 0$ ) and (critical dealers for  $c > 0$ ) do
13:         $route.append(current)$ 
14:        for dealer  $j \in N_{sorted}$  (by  $A_{current,j}$ ) do
15:          if  $j$  is critical then
16:            if ( $available\_capacity < \text{cars needed for } j$ ) and ( $route[-1] \neq 0$ ) then
17:              if remaining capacity for  $route[-1] > available\_capacity$  then
18:                Drop off  $available\_capacity$  at node  $route[-1]$ 
19:              end if
20:               $available\_capacity = 0$ 
21:              break
22:            end if
23:             $route\_time = route\_time + A_{current,j}$ 
24:             $current = j$ 
25:             $available\_capacity = available\_capacity - \text{cars needed for } j$ 
26:            Drop off needed cars at node  $j$ 
27:          end if
28:        end for
29:      end while
30:      if  $available\_capacity \neq 0$  then
31:        if remaining capacity for  $route[-1] > available\_capacity$  then
32:          Drop off  $available\_capacity$  at node  $route[-1]$ 
33:        end if
34:      end if
35:       $route.append(0)$ 
36:       $route\_time = route\_time + A_{current,0}$ 
37:      Lock truck  $k$  for  $route\_time$  days
38:    end while
39:  end for
40: end for
```

Figure 1.1: Console Outputs of the Gurobi Solver

Heuristic Solution Plots

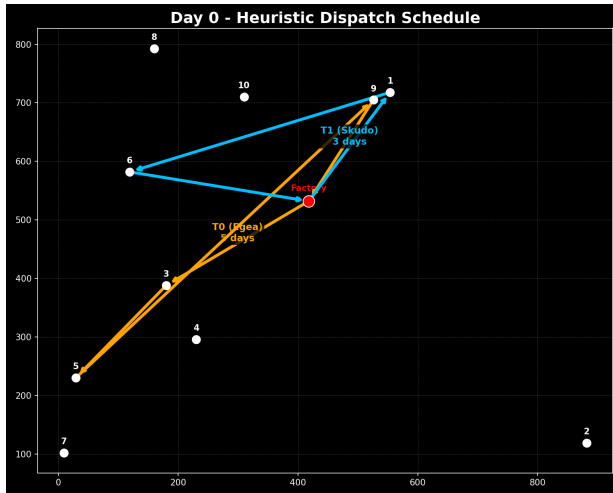


Figure 4.1: Routing Schedule on Day 0

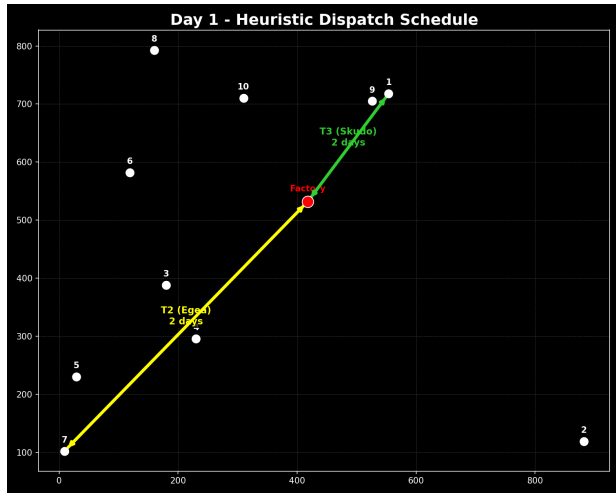


Figure 4.2: Routing Schedule on Day 1

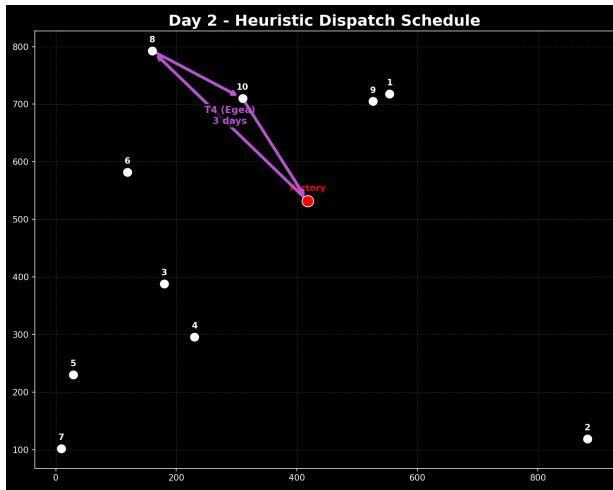


Figure 4.3: Routing Schedule on Day 2

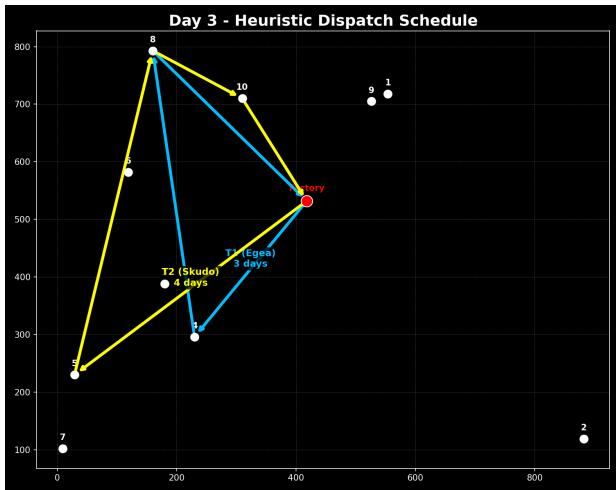


Figure 4.4: Routing Schedule on Day 3

Table 1.1: Backorder cost vs Backorder Amounts & Objective Value

B. Multiplier	Obj. Value	Backorder Cost	Unmet Units	Trans. Cost
0.5	18516.75	750.00	3	15226.71
1	19258.86	0	0	16118.82
2	23846.82	3999.99	4	17566.79
5	19720.89	0	0	17160.85

(Note: The 2.0x run exhibiting 4 unmet units is an anomaly caused by the time limit, the solver was terminated while exploring a branch).

Table 2.1: *Truck Numbers and Objective Value, Exact Solution*

# of Trucks	Obj. Value	Trans. Cost	Hold. Cost	Backorder Cost	Unsold Cost
4	29742.46	15862.42	1380.00	12500.00	0
6	19132.69	15992.65	3140.00	0	0
8	18157.38	15337.34	2820.00	0	0

Table: *Truck Numbers and Objective Value, Heuristic Solution*

# of Trucks	Obj. Value	Trans. Cost	Backorder Cost	Unsold Cost	Time
4	1122110.41	15610.41	12500.00	1095000.00	0.0004
6	1373188.63	23188.63	0	1350000.00	0.0004
8	1389096.47	24096.47	0	1365000.00	0.0005